```
C:\Users\huawei\PyCharmMiscProject\.venv\Scripts\python.exe D:\Final_Homework\main.py
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778429057.636659   21108 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778429059.810278   21108 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1778429080.328669   21108 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.

================================================================
纽约出租车数据问答系统
================================================================

正在加载数据...
正在构建需求模型...
正在训练预测模型...
系统初始化完成!


================================================================
数据清洗
================================================================

清洗前数据形状: 3,066,766 行 × 19 列

【步骤1: 去除缺失值】
关键列缺失情况:
  列 'tpep_pickup_datetime': 缺失0行 (0.00%)
  列 'tpep_dropoff_datetime': 缺失0行 (0.00%)
  列 'trip_distance': 缺失0行 (0.00%)
  列 'fare_amount': 缺失0行 (0.00%)
  列 'total_amount': 缺失0行 (0.00%)
  列 'passenger_count': 缺失71,743行 (2.34%)

  因关键列缺失值删除行数: 71,743
  删除后数据形状: 2,995,023 行 × 19 列

【步骤2: 去除业务逻辑异常值】
```

```
  删除行程距离≤0的数据: 38,367行
  删除费用≤0的数据: 22,475行
  删除乘客数≤0的数据: 49,953行
  删除下车时间≤上车时间的数据: 63行

  因业务逻辑异常删除行数: 110,858
  删除后数据形状: 2,884,165 行 × 19 列

【步骤3: 处理分类列异常值】
  删除RatecodeID异常值(99): 11,210行

  因分类列异常删除行数: 11,210

【步骤4: 去除重复值】
  移除重复行数: 0
  去重后数据形状: 2,872,955 行 × 19 列

======================================================================
数据清洗总结
======================================================================

原始数据行数: 3,066,766
清洗后数据行数: 2,872,955
总共删除行数: 193,811 (6.32%)
  - 因缺失值删除: 71,743行
  - 因业务逻辑异常删除: 110,858行
  - 因重复值删除: 0行

清洗后数据预览:
   VendorID tpep_pickup_datetime tpep_dropoff_datetime  passenger_count  trip_distance  RatecodeID store_and_fwd_flag  PULocationID  DOLocationID  payment_type  fare_amount  extra  mta_tax  tip_amount  tolls_amount  improvement_surcharge  total_amount  congestion_surcharge  airport_fee
0         2  2023-01-01 00:32:10   2023-01-01 00:40:36              1.0           0.97         1.0                  N           161           141             2          9.3    1.0      0.5        0.00           0.0                    1.0         14.30                   2.5          0.0
1         2  2023-01-01 00:55:08   2023-01-01 01:01:27              1.0           1.10         1.0                  N            43           237             1          7.9    1.0      0.5        4.00           0.0                    1.0         16.90                   2.5          0.0
2         2  2023-01-01 00:25:04   2023-01-01 00:37:49              1.0           2.51         1.0                  N            48           238             1         14.9    1.0      0.5       15.00           0.0                    1.0         34.90                   2.5          0.0
3         2  2023-01-01 00:10:29   2023-01-01 00:21:19              1.0           1.43         1.0                  N           107            79             1         11.4    1.0      0.5        3.28           0.0                    1.0         19.68                   2.5          0.0
4         2  2023-01-01 00:50:34   2023-01-01 01:02:52              1.0           1.84         1.0                  N           161           137             1         12.8    1.0      0.5       10.00           0.0                    1.0         27.80                   2.5          0.0
正在提取时间特征...
时间特征提取完成
```

```
【分析1: 出行需求时间规律】
 □ 01_hourly_demand.png 已保存
 □ 01_weekday_hour_heatmap.png 已保存

【分析2: 区域热度分析】
 □ 02_top10_locations.png 已保存
 □ 02_top5_hourly.png 已保存
 □ 02_od_flows.png 已保存

【分析3: 车费影响因素分析】
 数据量较大，已采样至100,000条用于绘图
 □ 03_distance_fare.png 已保存
 □ 03_hourly_fare.png 已保存
 □ 03_passenger_fare.png 已保存

【分析4: 支付方式与费用结构深度洞察】
 □ 04_payment_analysis.png 已保存
 □ 04_fee_structure.png 已保存
 □ 04_hourly_payment_preference.png 已保存

【步骤1: 数据加载与特征工程】
正在提取时间特征...
时间特征提取完成

【步骤2: 构建目标变量】
聚合后数据形状: (18212, 11)
区域数量: 250
需求量范围: 1 - 2723
平均需求量: 157.75

【步骤3: 准备训练数据】
训练集大小: 14,569
测试集大小: 3,643

【步骤4: 构建神经网络模型】
神经网络结构:
Model: "sequential"
┌─────────────────────────────────┬────────────────────────┬───────────────┐
│ Layer (type)                    │ Output Shape           │       Param # │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dense (Dense)                   │ (None, 128)            │         1,024 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ batch_normalization             │ (None, 128)            │           512 │
│ (BatchNormalization)            │                        │               │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dropout (Dropout)               │ (None, 128)            │             0 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dense_1 (Dense)                 │ (None, 64)             │         8,256 │
```

```
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ batch_normalization_1           │ (None, 64)             │           256 │
│ (BatchNormalization)            │                        │               │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dropout_1 (Dropout)             │ (None, 64)             │             0 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dense_2 (Dense)                 │ (None, 32)             │         2,080 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ batch_normalization_2           │ (None, 32)             │           128 │
│ (BatchNormalization)            │                        │               │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dropout_2 (Dropout)             │ (None, 32)             │             0 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dense_3 (Dense)                 │ (None, 1)              │            33 │
└─────────────────────────────────┴────────────────────────┴───────────────┘
 Total params: 12,289 (48.00 KB)
 Trainable params: 11,841 (46.25 KB)
 Non-trainable params: 448 (1.75 KB)

【步骤5: 训练神经网络】
Epoch 1/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 2s 7ms/step - loss: 1.9161 - mae: 1.0035 - val_loss: 1.0512 - val_mae: 0.6198 - learning_rate: 0.0010
Epoch 2/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1.3901 - mae: 0.8196 - val_loss: 1.0371 - val_mae: 0.6042 - learning_rate: 0.0010
Epoch 3/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1.2757 - mae: 0.7672 - val_loss: 1.0246 - val_mae: 0.6120 - learning_rate: 0.0010
Epoch 4/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1.2282 - mae: 0.7413 - val_loss: 1.0143 - val_mae: 0.6088 - learning_rate: 0.0010
Epoch 5/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1.1906 - mae: 0.7302 - val_loss: 1.0130 - val_mae: 0.6045 - learning_rate: 0.0010
Epoch 6/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1.1452 - mae: 0.7078 - val_loss: 1.0032 - val_mae: 0.6083 - learning_rate: 0.0010
Epoch 7/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 1.1221 - mae: 0.6983 - val_loss: 0.9959 - val_mae: 0.6136 - learning_rate: 0.0010
Epoch 8/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1.1102 - mae: 0.6889 - val_loss: 0.9915 - val_mae: 0.6141 - learning_rate: 0.0010
Epoch 9/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1.0774 - mae: 0.6763 - val_loss: 0.
```

```
9858 - val_mae: 0.6126 - learning_rate: 0.0010
Epoch 10/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 1.0704 - mae: 0.6722 - val_loss: 0.9841 - val_mae: 0.6151 - learning_rate: 0.0010
Epoch 11/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 1.0623 - mae: 0.6685 - val_loss: 0.9813 - val_mae: 0.6163 - learning_rate: 0.0010
Epoch 12/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1.0539 - mae: 0.6646 - val_loss: 0.9792 - val_mae: 0.6172 - learning_rate: 0.0010
Epoch 13/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1.0430 - mae: 0.6586 - val_loss: 0.9755 - val_mae: 0.6165 - learning_rate: 0.0010
Epoch 14/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1.0383 - mae: 0.6554 - val_loss: 0.9737 - val_mae: 0.6138 - learning_rate: 0.0010
Epoch 15/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 1.0317 - mae: 0.6536 - val_loss: 0.9709 - val_mae: 0.6140 - learning_rate: 0.0010
Epoch 16/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1.0198 - mae: 0.6508 - val_loss: 0.9683 - val_mae: 0.6145 - learning_rate: 0.0010
Epoch 17/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1.0135 - mae: 0.6467 - val_loss: 0.9667 - val_mae: 0.6133 - learning_rate: 0.0010
Epoch 18/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1.0077 - mae: 0.6441 - val_loss: 0.9666 - val_mae: 0.6138 - learning_rate: 0.0010
Epoch 19/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 1.0057 - mae: 0.6424 - val_loss: 0.9640 - val_mae: 0.6135 - learning_rate: 0.0010
Epoch 20/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 1.0024 - mae: 0.6409 - val_loss: 0.9619 - val_mae: 0.6099 - learning_rate: 0.0010
Epoch 21/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9958 - mae: 0.6393 - val_loss: 0.9606 - val_mae: 0.6123 - learning_rate: 0.0010
Epoch 22/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.9890 - mae: 0.6344 - val_loss: 0.9595 - val_mae: 0.6133 - learning_rate: 0.0010
Epoch 23/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 0.9880 - mae: 0.6343 - val_loss: 0.9567 - val_mae: 0.6160 - learning_rate: 0.0010
Epoch 24/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9840 - mae: 0.6346 - val_loss: 0.9541 - val_mae: 0.6153 - learning_rate: 0.0010
Epoch 25/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9812 - mae: 0.6330 - val_loss: 0.9515 - val_mae: 0.6151 - learning_rate: 0.0010
Epoch 26/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.9740 - mae: 0.6311 - val_loss: 0.9504 - val_mae: 0.6122 - learning_rate: 0.0010
Epoch 27/100
```

```
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.9735 - mae: 0.6307 - val_loss: 0.9488 - val_mae: 0.6119 - learning_rate: 0.0010
Epoch 28/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9678 - mae: 0.6295 - val_loss: 0.9449 - val_mae: 0.6101 - learning_rate: 0.0010
Epoch 29/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9717 - mae: 0.6300 - val_loss: 0.9439 - val_mae: 0.6127 - learning_rate: 0.0010
Epoch 30/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9689 - mae: 0.6306 - val_loss: 0.9404 - val_mae: 0.6109 - learning_rate: 0.0010
Epoch 31/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9615 - mae: 0.6285 - val_loss: 0.9382 - val_mae: 0.6114 - learning_rate: 0.0010
Epoch 32/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9638 - mae: 0.6298 - val_loss: 0.9360 - val_mae: 0.6125 - learning_rate: 0.0010
Epoch 33/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9592 - mae: 0.6286 - val_loss: 0.9339 - val_mae: 0.6122 - learning_rate: 0.0010
Epoch 34/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9519 - mae: 0.6251 - val_loss: 0.9322 - val_mae: 0.6108 - learning_rate: 0.0010
Epoch 35/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9493 - mae: 0.6259 - val_loss: 0.9302 - val_mae: 0.6121 - learning_rate: 0.0010
Epoch 36/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9490 - mae: 0.6254 - val_loss: 0.9281 - val_mae: 0.6093 - learning_rate: 0.0010
Epoch 37/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9480 - mae: 0.6254 - val_loss: 0.9268 - val_mae: 0.6107 - learning_rate: 0.0010
Epoch 38/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9466 - mae: 0.6271 - val_loss: 0.9243 - val_mae: 0.6100 - learning_rate: 0.0010
Epoch 39/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9445 - mae: 0.6250 - val_loss: 0.9232 - val_mae: 0.6118 - learning_rate: 0.0010
Epoch 40/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.9402 - mae: 0.6248 - val_loss: 0.9210 - val_mae: 0.6134 - learning_rate: 0.0010
Epoch 41/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9426 - mae: 0.6256 - val_loss: 0.9185 - val_mae: 0.6132 - learning_rate: 0.0010
Epoch 42/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9394 - mae: 0.6236 - val_loss: 0.9171 - val_mae: 0.6141 - learning_rate: 0.0010
Epoch 43/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.9375 - mae: 0.6245 - val_loss: 0.9138 - val_mae: 0.6145 - learning_rate: 0.0010
Epoch 44/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9340 - mae: 0.6231 - val_loss: 0.9127 - val_mae: 0.6110 - learning_rate: 0.0010
```

```
Epoch 45/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9316 - mae: 0.6221 - val_loss: 0.9092 - val_mae: 0.6112 - learning_rate: 0.0010
Epoch 46/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9299 - mae: 0.6226 - val_loss: 0.9080 - val_mae: 0.6088 - learning_rate: 0.0010
Epoch 47/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9276 - mae: 0.6210 - val_loss: 0.9077 - val_mae: 0.6118 - learning_rate: 0.0010
Epoch 48/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9289 - mae: 0.6235 - val_loss: 0.9063 - val_mae: 0.6117 - learning_rate: 0.0010
Epoch 49/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.9230 - mae: 0.6214 - val_loss: 0.9047 - val_mae: 0.6100 - learning_rate: 0.0010
Epoch 50/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9259 - mae: 0.6234 - val_loss: 0.9041 - val_mae: 0.6125 - learning_rate: 0.0010
Epoch 51/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9197 - mae: 0.6209 - val_loss: 0.9022 - val_mae: 0.6116 - learning_rate: 0.0010
Epoch 52/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9228 - mae: 0.6234 - val_loss: 0.9012 - val_mae: 0.6121 - learning_rate: 0.0010
Epoch 53/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9182 - mae: 0.6215 - val_loss: 0.8993 - val_mae: 0.6096 - learning_rate: 0.0010
Epoch 54/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9179 - mae: 0.6204 - val_loss: 0.8982 - val_mae: 0.6084 - learning_rate: 0.0010
Epoch 55/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9183 - mae: 0.6221 - val_loss: 0.8963 - val_mae: 0.6101 - learning_rate: 0.0010
Epoch 56/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.9191 - mae: 0.6225 - val_loss: 0.8943 - val_mae: 0.6091 - learning_rate: 0.0010
Epoch 57/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9123 - mae: 0.6199 - val_loss: 0.8928 - val_mae: 0.6110 - learning_rate: 0.0010
Epoch 58/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9137 - mae: 0.6194 - val_loss: 0.8911 - val_mae: 0.6100 - learning_rate: 0.0010
Epoch 59/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.9116 - mae: 0.6198 - val_loss: 0.8904 - val_mae: 0.6085 - learning_rate: 0.0010
Epoch 60/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.9111 - mae: 0.6201 - val_loss: 0.8902 - val_mae: 0.6061 - learning_rate: 0.0010
Epoch 61/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.9105 - mae: 0.6198 - val_loss: 0.8868 - val_mae: 0.6091 - learning_rate: 0.0010
Epoch 62/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 0.9075 - mae: 0.6189 - val_loss: 0.
```

```
8859 - val_mae: 0.6062 - learning_rate: 0.0010
Epoch 63/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.9058 - mae: 0.6184 - val_loss: 0.8845 - val_mae: 0.6074 - learning_rate: 0.0010
Epoch 64/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9060 - mae: 0.6195 - val_loss: 0.8842 - val_mae: 0.6104 - learning_rate: 0.0010
Epoch 65/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9073 - mae: 0.6207 - val_loss: 0.8811 - val_mae: 0.6109 - learning_rate: 0.0010
Epoch 66/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9040 - mae: 0.6181 - val_loss: 0.8800 - val_mae: 0.6126 - learning_rate: 0.0010
Epoch 67/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9041 - mae: 0.6184 - val_loss: 0.8808 - val_mae: 0.6145 - learning_rate: 0.0010
Epoch 68/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.8978 - mae: 0.6172 - val_loss: 0.8786 - val_mae: 0.6104 - learning_rate: 0.0010
Epoch 69/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8995 - mae: 0.6176 - val_loss: 0.8788 - val_mae: 0.6117 - learning_rate: 0.0010
Epoch 70/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.9000 - mae: 0.6191 - val_loss: 0.8793 - val_mae: 0.6116 - learning_rate: 0.0010
Epoch 71/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8990 - mae: 0.6167 - val_loss: 0.8778 - val_mae: 0.6098 - learning_rate: 0.0010
Epoch 72/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.8944 - mae: 0.6149 - val_loss: 0.8740 - val_mae: 0.6071 - learning_rate: 0.0010
Epoch 73/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8928 - mae: 0.6156 - val_loss: 0.8744 - val_mae: 0.6088 - learning_rate: 0.0010
Epoch 74/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8950 - mae: 0.6163 - val_loss: 0.8727 - val_mae: 0.6054 - learning_rate: 0.0010
Epoch 75/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.8947 - mae: 0.6157 - val_loss: 0.8724 - val_mae: 0.6088 - learning_rate: 0.0010
Epoch 76/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8930 - mae: 0.6152 - val_loss: 0.8719 - val_mae: 0.6079 - learning_rate: 0.0010
Epoch 77/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8912 - mae: 0.6155 - val_loss: 0.8697 - val_mae: 0.6042 - learning_rate: 0.0010
Epoch 78/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8891 - mae: 0.6140 - val_loss: 0.8707 - val_mae: 0.6068 - learning_rate: 0.0010
Epoch 79/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.8857 - mae: 0.6132 - val_loss: 0.8666 - val_mae: 0.6098 - learning_rate: 0.0010
Epoch 80/100
```

```
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8887 - mae: 0.6134 - val_loss: 0.8687 - val_mae: 0.6118 - learning_rate: 0.0010
Epoch 81/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8884 - mae: 0.6133 - val_loss: 0.8660 - val_mae: 0.6100 - learning_rate: 0.0010
Epoch 82/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8862 - mae: 0.6137 - val_loss: 0.8642 - val_mae: 0.6127 - learning_rate: 0.0010
Epoch 83/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.8868 - mae: 0.6129 - val_loss: 0.8635 - val_mae: 0.6119 - learning_rate: 0.0010
Epoch 84/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8781 - mae: 0.6098 - val_loss: 0.8589 - val_mae: 0.6163 - learning_rate: 0.0010
Epoch 85/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8803 - mae: 0.6094 - val_loss: 0.8594 - val_mae: 0.6181 - learning_rate: 0.0010
Epoch 86/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.8760 - mae: 0.6086 - val_loss: 0.8495 - val_mae: 0.6123 - learning_rate: 0.0010
Epoch 87/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8721 - mae: 0.6075 - val_loss: 0.8436 - val_mae: 0.6071 - learning_rate: 0.0010
Epoch 88/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8717 - mae: 0.6059 - val_loss: 0.8421 - val_mae: 0.6100 - learning_rate: 0.0010
Epoch 89/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.8656 - mae: 0.6036 - val_loss: 0.8352 - val_mae: 0.6060 - learning_rate: 0.0010
Epoch 90/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.8589 - mae: 0.6008 - val_loss: 0.8196 - val_mae: 0.5938 - learning_rate: 0.0010
Epoch 91/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8499 - mae: 0.5996 - val_loss: 0.8193 - val_mae: 0.5887 - learning_rate: 0.0010
Epoch 92/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8407 - mae: 0.5958 - val_loss: 0.8122 - val_mae: 0.5764 - learning_rate: 0.0010
Epoch 93/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8417 - mae: 0.5942 - val_loss: 0.8021 - val_mae: 0.5658 - learning_rate: 0.0010
Epoch 94/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8346 - mae: 0.5902 - val_loss: 0.7937 - val_mae: 0.5570 - learning_rate: 0.0010
Epoch 95/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8202 - mae: 0.5850 - val_loss: 0.7973 - val_mae: 0.5514 - learning_rate: 0.0010
Epoch 96/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.8198 - mae: 0.5869 - val_loss: 0.7972 - val_mae: 0.5443 - learning_rate: 0.0010
Epoch 97/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.8127 - mae: 0.5792 - val_loss: 0.7841 - val_mae: 0.5436 - learning_rate: 0.0010
```

```
Epoch 98/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.8127 - mae: 0.5828 - val_loss: 0.7817 - val_mae: 0.5545 - learning_rate: 0.0010
Epoch 99/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 0.8120 - mae: 0.5814 - val_loss: 0.7804 - val_mae: 0.5568 - learning_rate: 0.0010
Epoch 100/100
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 0.8115 - mae: 0.5793 - val_loss: 0.7724 - val_mae: 0.5476 - learning_rate: 0.0010
Restoring model weights from the end of the best epoch: 100.

【步骤6: 绘制Loss曲线】
  □ nn_training_curves.png 已保存

【步骤7: 神经网络模型评估】
114/114 ━━━━━━━━━━━━━━━━━━━━ 0s 1ms/step
神经网络测试结果:
  MAE:  156.7630
  RMSE: 245.8496
  □ nn_predictions.png 已保存

【步骤8: 训练随机森林模型】
训练随机森林中...
随机森林训练完成

【步骤9: 随机森林模型评估】
随机森林测试结果:
  MAE:  31.0641
  RMSE: 73.1318
  □ rf_predictions.png 已保存

============================================================
模型对比分析
============================================================
  □ model_comparison.png 已保存
  □ feature_importance.png 已保存
  □ residuals_comparison.png 已保存

============================================================
模型对比总结
============================================================

指标          神经网络        随机森林        较优模型
------------------------------------------------------------
MAE           156.7630        31.0641         随机森林
RMSE          245.8496        73.1318         随机森林

性能差异:
  MAE差异: 80.18%
```

```
  RMSE差异: 70.25%

分析方法优劣:
----------------------------------------
神经网络优势:
  1. 能捕捉复杂的非线性关系
  2. 通过循环特征更好地理解时间的周期性
  3. 可扩展性强，可添加更多特征
  4. 适合大规模数据训练

随机森林优势:
  1. 训练速度快，不需要特征标准化
  2. 可解释性强，能输出特征重要性
  3. 对超参数不那么敏感
  4. 小数据集上表现更稳定
列名: ['VendorID', 'tpep_pickup_datetime', 'tpep_dropoff_datetime', 'passenger_count', 'trip_distance', 'RatecodeID', 'store_and_fwd_flag', 'PULocationID', 'DOLocationID', 'payment_type', 'fare_amount', 'extra', 'mta_tax', 'tip_amount', 'tolls_amount', 'improvement_surcharge', 'total_amount', 'congestion_surcharge', 'airport_fee', 'pickup_datetime', 'pickup_hour', 'pickup_weekday', 'pickup_day', 'is_weekend', 'is_rush_hour', 'hour_sin', 'hour_cos', 'weekday_sin', 'weekday_cos']
is_weekend 是否存在: True
is_weekend 类型: bool
is_weekend 唯一值: [ True False]

============================================================
欢迎使用纽约出租车数据问答系统！
============================================================

支持的问题类型：
  1. 时段查询  - '区域100 工作日分时段有多少订单'
  2. 区域排名  - '上车量排名前10的区域'
  3. 需求预测  - '预测区域100 周一下午5点需求量'
  4. 路线查询  - '从区域100到区域200的费用'
  5. 支付分析  - '各种支付方式的小费对比'
  6. 高峰分析  - '周五哪些时段最繁忙'

输入 'quit' 或 'exit' 退出系统
------------------------------------------------------------

请输入您的问题：上车量排名前10的区域

□ 区域热度排名（上车量 TOP 10）：
   1. 区域132：152,121单
   2. 区域237：141,089单
   3. 区域236：130,857单
   4. 区域161：129,050单
   5. 区域186：104,779单
   6. 区域162：100,787单
   7. 区域142：94,861单
   8. 区域230：94,212单
```

```
  9. 区域138：86,750单
  10. 区域170：83,933单
  □ 图表已保存：outputs/qa_location_ranking.png
------------------------------------------------------------

请输入您的问题：quit

感谢使用，再见！

进程已结束，退出代码为 0
```